

Name/Surname:

Ammaarah Khan

Student Number:

ST10447616

Module Name:

Programming 3A

Module Code:

PROG7311

Assessment Type:

POE Part 02

Date:

22 April 2026

LINKS:

GitHub Repository Link:

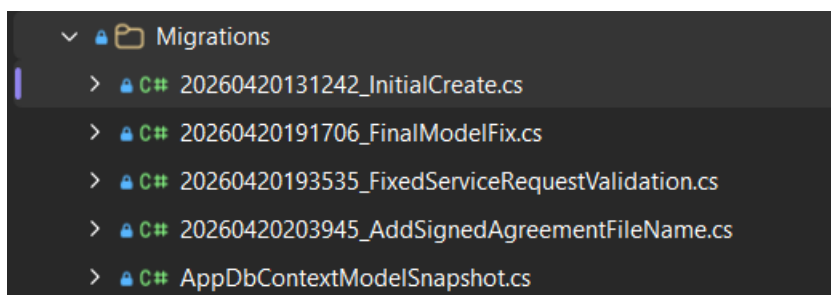
<https://github.com/EMKNPM/part-2-the-core-prototype-unit-testing-ammaarahk05.git>

YouTube Link:

<https://www.youtube.com/watch?v=akytQ40jd2U>

1. Database Migration Scripts:

1st Migration: InitialCreate.cs



migration creating the Clients table and the

Contracts table:

```

migrationBuilder.CreateTable(
    name: "Clients",
    columns: table => new
    {
        Id = table.Column<int>(type: "int", nullable: false)
            .Annotation("SqlServer:Identity", "1, 1"),
        Name = table.Column<string>(type: "nvarchar(max)", nullable: false),
        ContactDetails = table.Column<string>(type: "nvarchar(max)", nullable: false),
        Region = table.Column<string>(type: "nvarchar(max)", nullable: false)
    },
    constraints: table =>
    {
        table.PrimaryKey("PK_Clients", x => x.Id);
    });

migrationBuilder.CreateTable(
    name: "Contracts",
    columns: table => new
    {
        Id = table.Column<int>(type: "int", nullable: false)
            .Annotation("SqlServer:Identity", "1, 1"),
        StartDate = table.Column<DateTime>(type: "datetime2", nullable: false),
        EndDate = table.Column<DateTime>(type: "datetime2", nullable: false),
        Status = table.Column<string>(type: "nvarchar(max)", nullable: false),
        ServiceLevel = table.Column<string>(type: "nvarchar(max)", nullable: false),
        ClientId = table.Column<int>(type: "int", nullable: false),
        SignedAgreementPath = table.Column<string>(type: "nvarchar(max)", nullable: false)
    },
    constraints: table =>
    {
        table.PrimaryKey("PK_Contracts", x => x.Id);
        table.ForeignKey(
            name: "FK_Contracts_Clients_ClientId",
            column: x => x.ClientId,
            principalTable: "Clients",
            principalColumn: "Id",
            onDelete: ReferentialAction.Cascade);
    });

```

migration creating the ServiceRequests table:

```

migrationBuilder.CreateTable(
    name: "ServiceRequests",
    columns: table => new
    {
        Id = table.Column<int>(type: "int", nullable: false)
            .Annotation("SqlServer:Identity", "1, 1"),
        Description = table.Column<string>(type: "nvarchar(max)", nullable: false),
        Cost = table.Column<decimal>(type: "decimal(18,2)", precision: 18, scale: 2, nullable: false),
        Status = table.Column<string>(type: "nvarchar(max)", nullable: false),
        ContractId = table.Column<int>(type: "int", nullable: false)
    },
    constraints: table =>
    {
        table.PrimaryKey("PK_ServiceRequests", x => x.Id);
        table.ForeignKey(
            name: "FK_ServiceRequests_Contracts_ContractId",
            column: x => x.ContractId,
            principalTable: "Contracts",
            principalColumn: "Id",
            onDelete: ReferentialAction.Cascade);
    });

```

2nd Migration – FinalModelFix.cs (Rollback logic):

```

protected override void Down(MigrationBuilder migrationBuilder)
{
    migrationBuilder.AlterColumn<string>(
        name: "Status",
        table: "ServiceRequests",
        type: "nvarchar(max)",
        nullable: false,
        defaultValue: "",
        oldClrType: typeof(string),
        oldType: "nvarchar(max)",
        oldNullable: true);

    migrationBuilder.AlterColumn<string>(
        name: "SignedAgreementPath",
        table: "Contracts",
        type: "nvarchar(max)",
        nullable: false,
        defaultValue: "",
        oldClrType: typeof(string),
        oldType: "nvarchar(max)",
        oldNullable: true);

    migrationBuilder.AlterColumn<string>(
        name: "ServiceLevel",
        table: "Contracts",
        type: "nvarchar(max)",
        nullable: false,
        defaultValue: "",
        oldClrType: typeof(string),
        oldType: "nvarchar(max)",
        oldNullable: true);

    migrationBuilder.AlterColumn<string>(
        name: "Region",
        table: "Clients",
        type: "nvarchar(max)",
        nullable: false,
        defaultValue: "",
        oldClrType: typeof(string),
        oldType: "nvarchar(max)",
        oldNullable: true);

    migrationBuilder.AlterColumn<string>(
        name: "Name",
        table: "Clients",
        type: "nvarchar(max)",
        nullable: false,
        oldClrType: typeof(string),
        oldType: "nvarchar(100)",
        oldMaxLength: 100);

    migrationBuilder.AlterColumn<string>(
        name: "ContactDetails",
        table: "Clients",
        type: "nvarchar(max)",
        nullable: false,
        defaultValue: "",
        oldClrType: typeof(string),
        oldType: "nvarchar(max)",
        oldNullable: true);
}

```

3rd Migration(Fixed Service Request Validation):

```

namespace TechMoveLogisticSystem.Migrations
{
    /// <inheritdoc />
    1 reference
    public partial class FixedServiceRequestValidation : Migration
    {
        /// <inheritdoc />
        0 references
        protected override void Up(MigrationBuilder migrationBuilder)
        {
        }

        /// <inheritdoc />
        0 references
        protected override void Down(MigrationBuilder migrationBuilder)
        {
        }
    }
}

```

4th Migration (AddSignedAgreementFileName):

```

using Microsoft.EntityFrameworkCore.Migrations;

#nullable disable

namespace TechMoveLogisticSystem.Migrations
{
    /// <inheritdoc />
    1 reference
    public partial class AddSignedAgreementFileName : Migration
    {
        /// <inheritdoc />
        0 references
        protected override void Up(MigrationBuilder migrationBuilder)
        {
            migrationBuilder.AddColumn<string>(
                name: "SignedAgreementFileName",
                table: "Contracts",
                type: "nvarchar(max)",
                nullable: true);
        }

        /// <inheritdoc />
        0 references
        protected override void Down(MigrationBuilder migrationBuilder)
        {
            migrationBuilder.DropColumn(
                name: "SignedAgreementFileName",
                table: "Contracts");
        }
    }
}

```

AppDbContextModelSnapshot.cs

final model contains the required tables and fields:

```
modelBuilder.Entity("TechMoveLogisticSystem.Models.Client", b =>
{
    b.Property<int>("Id")
        .ValueGeneratedOnAdd()
        .HasColumnType("int");

    SqlServerPropertyBuilderExtensions.UseIdentityColumn(b.Property<int>("Id"));

    b.Property<string>("ContactDetails")
        .HasColumnType("nvarchar(max)");

    b.Property<string>("Name")
        .IsRequired()
        .HasMaxLength(100)
        .HasColumnType("nvarchar(100)");

    b.Property<string>("Region")
        .HasColumnType("nvarchar(max)");

    b.HasKey("Id");

    b.ToTable("Clients");
});
```

```
modelBuilder.Entity("TechMoveLogisticSystem.Models.Contract", b =>
{
    b.Property<int>("Id")
        .ValueGeneratedOnAdd()
        .HasColumnType("int");

    SqlServerPropertyBuilderExtensions.UseIdentityColumn(b.Property<int>("Id"));

    b.Property<int>("ClientId")
        .HasColumnType("int");

    b.Property<DateTime>("EndDate")
        .HasColumnType("datetime2");

    b.Property<string>("ServiceLevel")
        .HasColumnType("nvarchar(max)");

    b.Property<string>("SignedAgreementFileName")
        .HasColumnType("nvarchar(max)");

    b.Property<string>("SignedAgreementPath")
        .HasColumnType("nvarchar(max)");

    b.Property<DateTime>("StartDate")
        .HasColumnType("datetime2");

    b.Property<string>("Status")
        .IsRequired()
        .HasColumnType("nvarchar(max)");

    b.HasKey("Id");

    b.HasIndex("ClientId");

    b.ToTable("Contracts");
});
```

```

modelBuilder.Entity("TechMoveLogisticSystem.Models.ServiceRequest", b =>
{
    b.Property<int>("Id")
        .ValueGeneratedOnAdd()
        .HasColumnType("int");

    SqlServerPropertyBuilderExtensions.UseIdentityColumn(b.Property<int>("Id"));

    b.Property<int>("ContractId")
        .HasColumnType("int");

    b.Property<decimal>("Cost")
        .HasPrecision(18, 2)
        .HasColumnType("decimal(18,2)");

    b.Property<string>("Description")
        .IsRequired()
        .HasColumnType("nvarchar(max)");

    b.Property<string>("Status")
        .HasColumnType("nvarchar(max)");

    b.HasKey("Id");

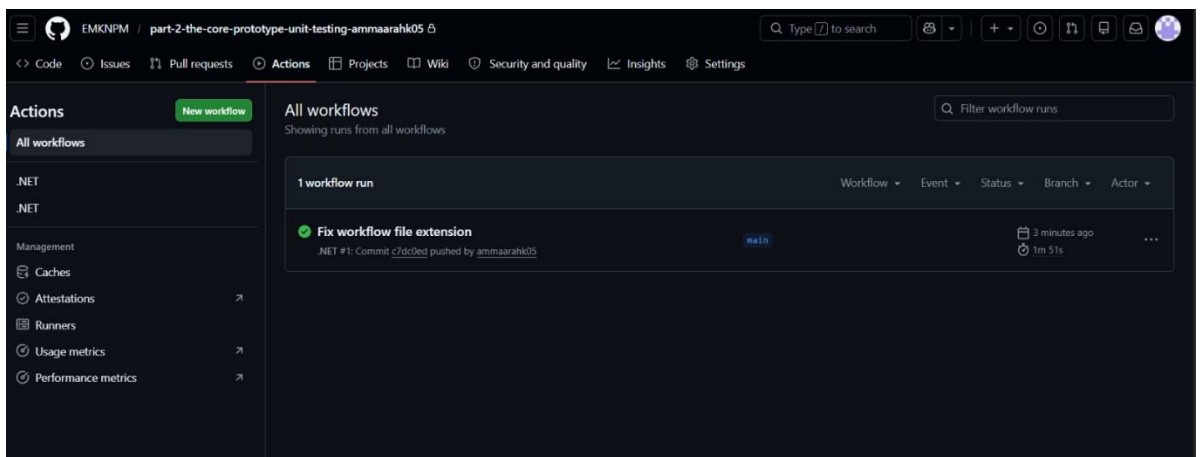
    b.HasIndex("ContractId");

    b.ToTable("ServiceRequests");
});

```

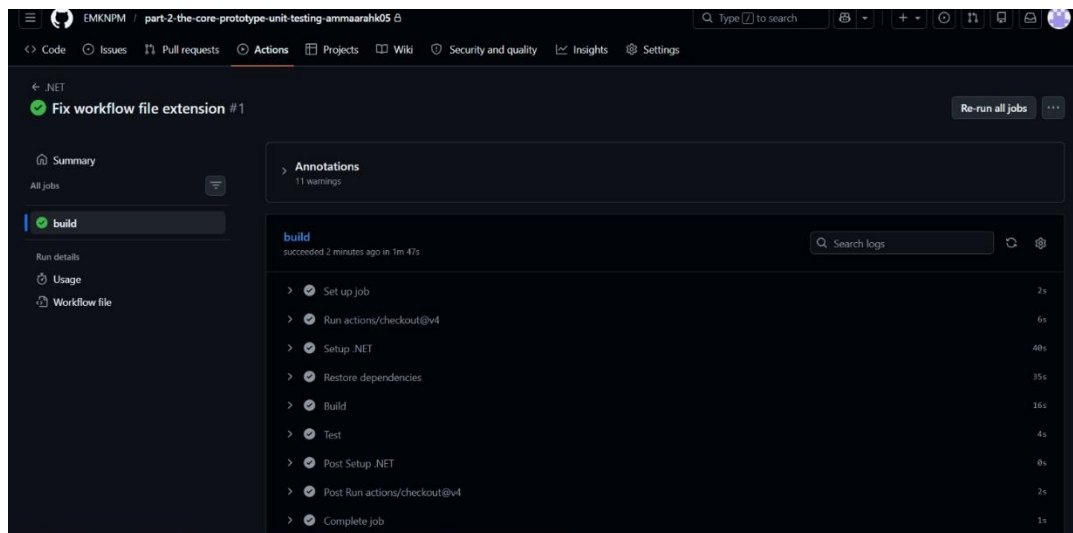
2. Test Execution Screenshots:

Workflow GitHub Actions – run successful



The screenshot shows the GitHub Actions interface for a repository named 'EMKNPM / part-2-the-core-prototype-unit-testing-ammaarahk05'. The 'Actions' tab is selected, displaying a list of workflow runs. A single workflow run is shown, titled 'Fix workflow file extension', which is marked as successful with a green checkmark. The run details indicate it was triggered by a commit push and completed successfully 3 minutes ago. The left sidebar shows the repository structure, including 'All workflows', '.NET', and 'Management'.

Build Successful:



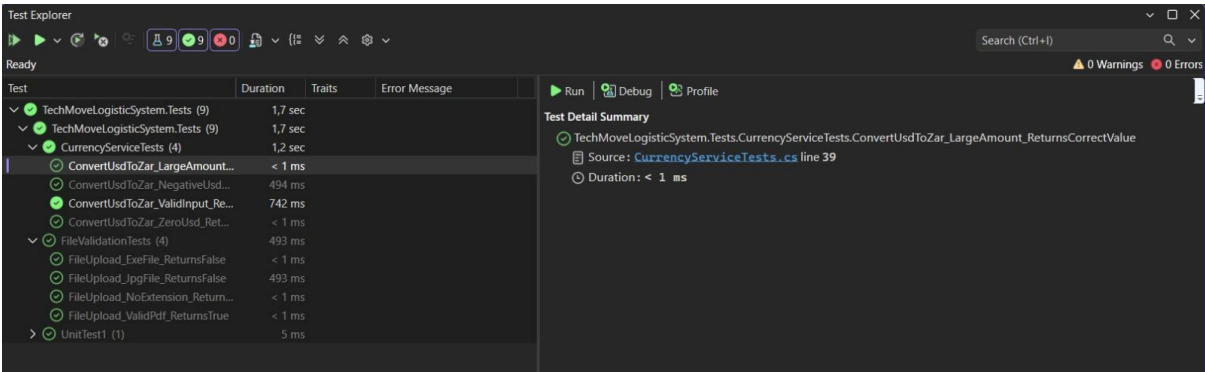
GitHub Actions > Build > Test results showing all Passed:

Test

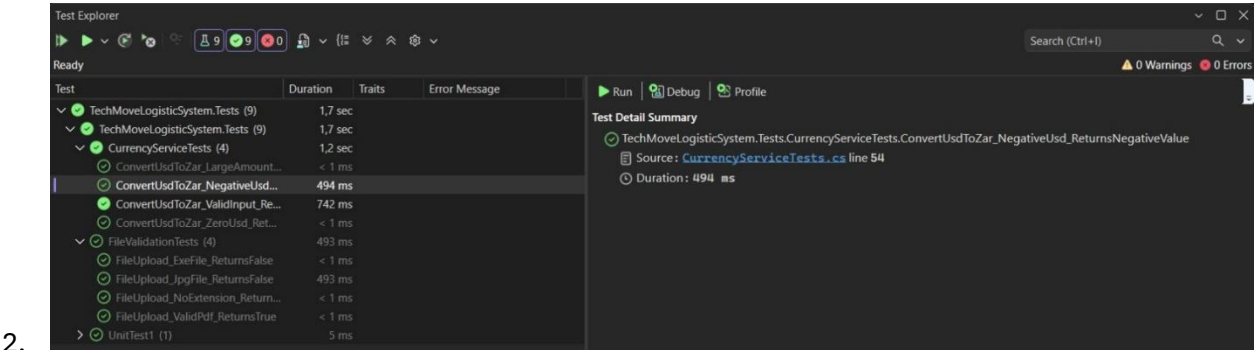
4s

```
1 ▶ Run dotnet test --no-build --verbosity normal
6 Build started 4/22/2026 5:19:33 PM.
7   1>Project "D:\a\part-2-the-core-prototype-unit-testing-ammaarahk05\part-2-the-core-prototype-unit-testing-ammaarahk05\TechMoveLogisticSystem.slnx"
   on node 1 (VSTest target(s)).
8   1>ValidateSolutionConfiguration:
9     Building solution configuration "Debug|Any CPU".
10  Test run for D:\a\part-2-the-core-prototype-unit-testing-ammaarahk05\part-2-the-core-prototype-unit-testing-
   ammaarahk05\TechMoveLogisticSystem.Tests\bin\Debug\net10.0\TechMoveLogisticSystem.Tests.dll (.NETCoreApp,Version=v10.0)
11  A total of 1 test files matched the specified pattern.
12  [xUnit.net 00:00:00.00] xUnit.net VSTest Adapter v3.1.4+50e68bbb8b (64-bit .NET 10.0.7)
13  [xUnit.net 00:00:00.10]   Discovering: TechMoveLogisticSystem.Tests
14  [xUnit.net 00:00:00.68]   Discovered: TechMoveLogisticSystem.Tests
15  [xUnit.net 00:00:00.72]   Starting: TechMoveLogisticSystem.Tests
16  [xUnit.net 00:00:00.79]   Finished: TechMoveLogisticSystem.Tests
17     Passed TechMoveLogisticSystem.Tests.UnitTest1.Test1 [3 ms]
18     Passed TechMoveLogisticSystem.Tests.FileValidationTests.FileUpload_JpgFile_ReturnsFalse [4 ms]
19     Passed TechMoveLogisticSystem.Tests.FileValidationTests.FileUpload_ExeFile_ReturnsFalse [< 1 ms]
20     Passed TechMoveLogisticSystem.Tests.FileValidationTests.FileUpload_ValidPdf_ReturnsTrue [< 1 ms]
21     Passed TechMoveLogisticSystem.Tests.FileValidationTests.FileUpload_NoExtension_ReturnsFalse [< 1 ms]
22     Passed TechMoveLogisticSystem.Tests.CurrencyServiceTests.ConvertUsdToZar_NegativeUsd_ReturnsNegativeValue [7 ms]
23     Passed TechMoveLogisticSystem.Tests.CurrencyServiceTests.ConvertUsdToZar_ValidInput_ReturnsCorrectValue [< 1 ms]
24     Passed TechMoveLogisticSystem.Tests.CurrencyServiceTests.ConvertUsdToZar_LargeAmount_ReturnsCorrectValue [< 1 ms]
25     Passed TechMoveLogisticSystem.Tests.CurrencyServiceTests.ConvertUsdToZar_ZeroUsd_ReturnsZero [< 1 ms]
26
27  Test Run Successful.
28  Total tests: 9
29     Passed: 9
30  Total time: 1.5567 Seconds
31   1>Done Building Project "D:\a\part-2-the-core-prototype-unit-testing-ammaarahk05\part-2-the-core-prototype-unit-testing-
   ammaarahk05\TechMoveLogisticSystem.slnx" (VSTest target(s)).
32
```

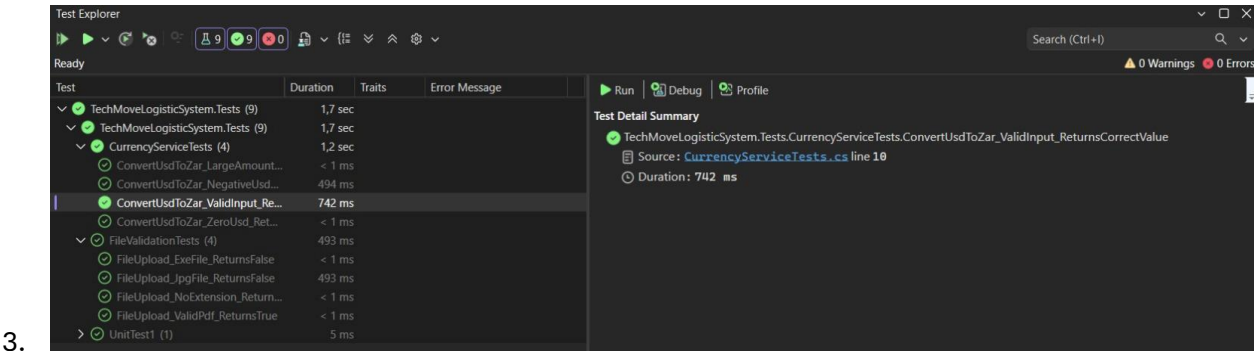
1. TEST 1: ConvertUsdToZar_LargeAmount_ReturnsCorrectValue

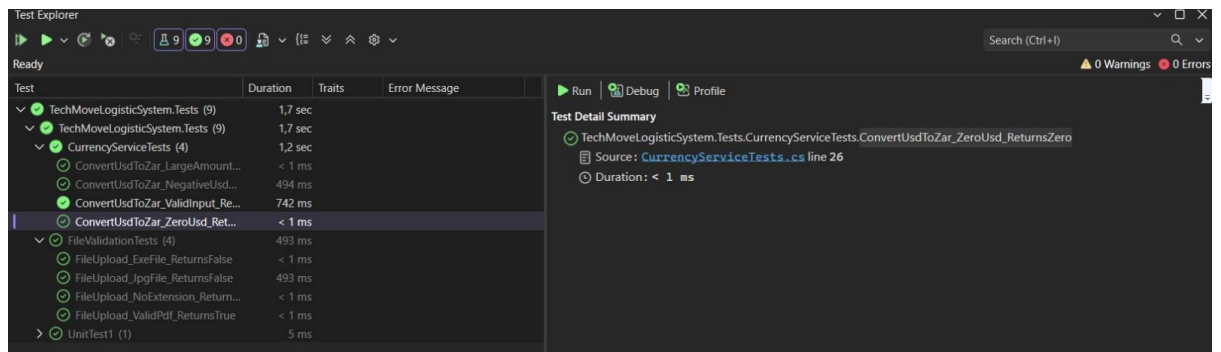


2. TEST 2:
ConvertUsdToZar_NegativeUsd_ReturnsNegativeValue

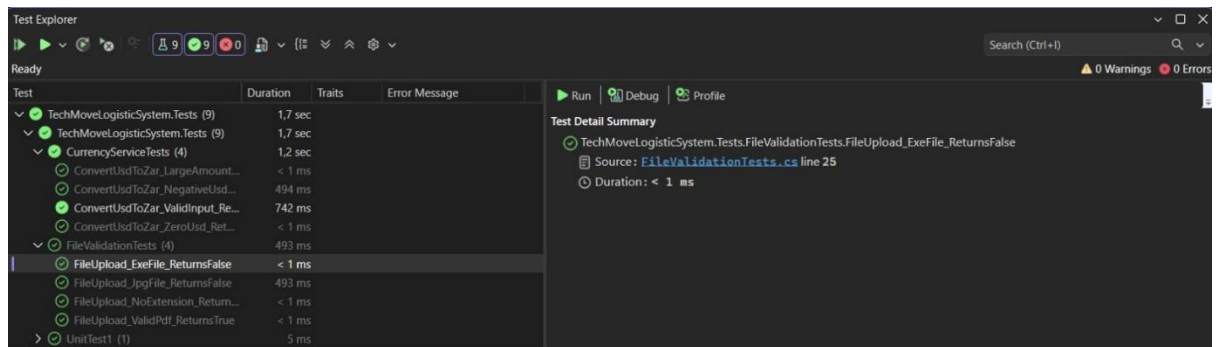


3. TEST 3:
ConvertUsdToZar_ValidInput_ReturnsCorrectValue

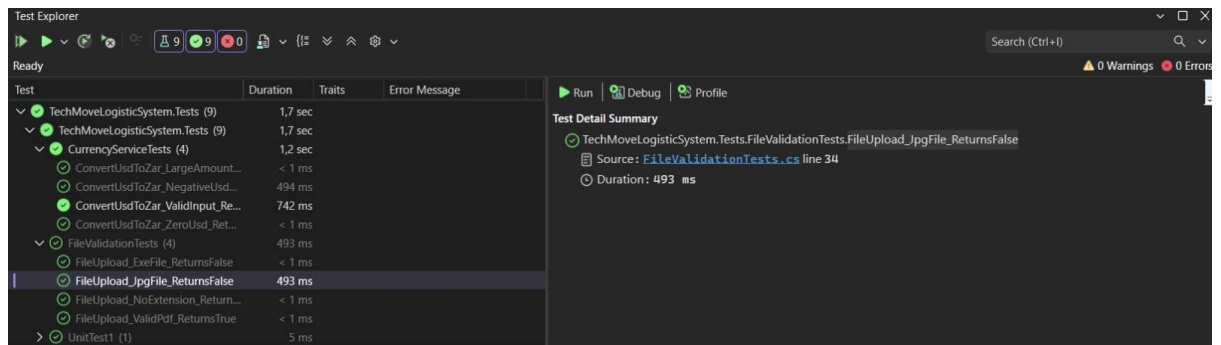




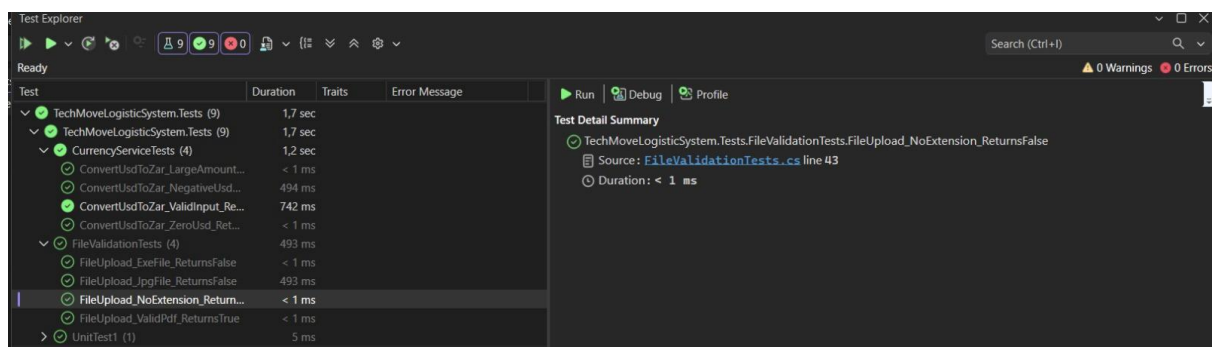
4. TEST 4:
ConvertUsdToZar_ZeroUsd_ReturnsZero



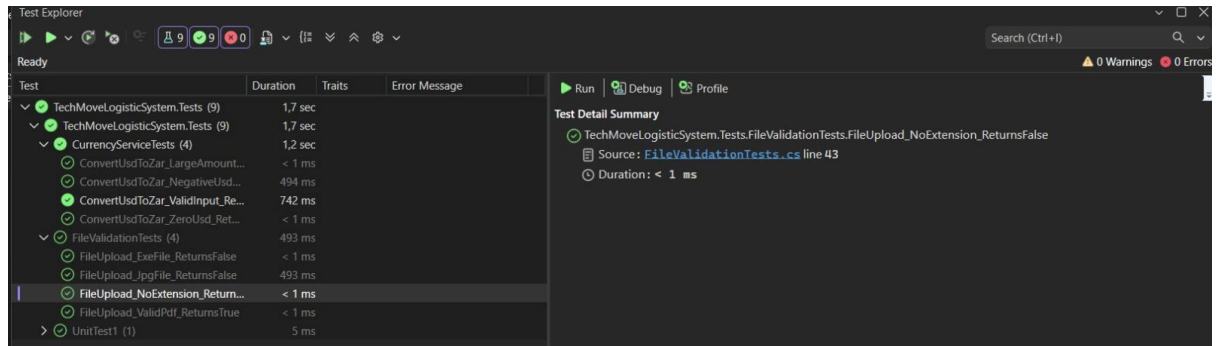
5. TEST 5:
FileUpload_ExeFile_ReturnsFalse



6. TEST 6:
FileUpload_JpgFile_ReturnsFalse



7. TEST 7:
FileUpload_NoExtension_ReturnsFalse



8.

TEST 8:

FileUpload_ValidPdf_ReturnsTrue